Weatherization Training Takes Off in the Southeast

To increase cooling efficiency in hot climates, trainers are taking the whole-house approach where it hasn't gone before.

by Bill Beachy and Alex Moore

A new training regimen is helping Mississippi adopt a whole-house approach to weatherization. And the whole-house approach is spreading to other hot-climate states. In 1995, representatives of the hot-climate states met in Dallas with building scientists from the nation's research laboratories to address what the states felt were inadequacies in the way weatherization was done in climates that were dominated by cooling energy needs. The southeastern states felt that national weatherization funding was primarily directed at reducing winter heating energy needs in cold climates, and that not enough attention was given to reducing cooling energy use in hot climates.

A result of the Dallas meeting was DOE's Hot Climate Initiative, the purpose of which was to help hot-climate states increase energy savings for their low-income clients. The initiative languished for several years due to lack of funding, but when funding became available again in 2001, the initiative was reinvigorated. As part of the initiative, D&R International developed a technical training project and piloted it in Mississippi in the summer of 2003. The focus of the training was to help the state adopt a whole-house approach to weatherization.

For phase 1 of the pilot, the organization that Bill Beachy heads, New River Center for Energy Research and Training (NRCERT), located in Christiansburg, Virginia, developed the curriculum and provided in-depth, hands-on weatherization training to Mississippi agencies and contractors. NRCERT trained the Mississippi agencies and contractors in diagnostic testing, health and safety testing, measure installation, and equipment use and maintenance.

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These Jackson, Mississippi, weatherization professionals learned the concepts of whole-house weatherization, and then put them into action.

While the training specifically addressed hot-climate issues, such as the installation and use of air conditioners, it did not target cooling measures alone. Mississippi agencies and contractors were trained to adopt a whole-house approach to weatherization—an approach that focuses on aligning and improving the air and thermal barriers of a house. The whole-house measures that reduce energy use for heating also cut cooling energy use. As a result, most of the Mississippi training is appropriate nationwide.

Phase 2 of the pilot provided additional training to a handful of agency and contractor personnel who demonstrated a command of phase 1. These individuals had taken the initiative to adopt the lessons learned in training in their local weatherization programs. By providing comprehensive training to this handful of self-starters, phase 2 of the pilot established a permanent core of knowledgeable weatherization professionals who will act as a technical resource for agencies and contractors statewide. This well-trained core will increase the likelihood that the whole-house approach will prosper in Mississippi.

"This training has really opened our eyes," says Ronza Anderson, Mississippi Weatherization program manager. "As a result, we're directing all agencies operating the Weatherization program to adopt the standards and techniques we learned from the trainers." While the phase 2 training was just recently completed, all indications are that the pilot has been successful. In fact, the pilot training is being replicated in Alabama in early 2004. The news is spreading around the region. At the request of North

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(left and right) A very dangerous disconnected furnace flue in a Mississippi home provided trainers with a dramatic teaching tool.

Carolina's Weatherization program, NRCERT recently began training North Carolina's local agencies and contractors using the Mississippi model. Tennessee has also requested the same training, which will begin in January.

Learning from Neighboring States

Before becoming the director of NRCERT, Bill Beachy was the weatherization program manager for the Commonwealth of Virginia. In 1991, we evaluated energy savings from typical weatherization measures that the state was using, such as replacing windows and doors. Virginia found that measures such as blower-door-directed air sealing offered a much greater savings potential—a finding that changed the direction of their program. Air sealing basements and attics, and using high-density wall insulation and other new measures increased space-heating savings by 10%–24% over savings realized by using typical measures. (For more on the Virginia evaluation, see "A Warm Wind Blows South: Virginia's Weatherization Evaluation," *HE* Jan/Feb '92, p. 15, and "Rehabilitating Virginia's Weatherization Program," *HE* Nov/Dec '92, p. 34.) Like Virginia, Mississippi has until recently emphasized replacing windows and doors. After hearing Virginia trainers share their first-hand experience, the Mississippi participants have begun to rethink their energy-saving measures.

Adopting a whole-house approach statewide can profoundly change the way the state, local agencies, and contractors do business.

The pilot in Mississippi led to three initial conclusions:

• Low-Income Home Energy Assistance Program (LIHEAP) funds for heating system replacement make health and safety efforts possible. The U.S. Department of Health and Human Services provides these funds.

• Technical training may be the easiest task in moving to a whole-house approach. The need for infrastructure—such as adequate equipment and the organization of agencies to purchase materials in bulk—may be the greatest obstacle to overcome.

• Adopting a whole-house approach represents a big change for some agencies. They should be given every opportunity to succeed. Some agencies that did not initially choose to take part in the new training or adapt to changing contracts could have lost grant money from the state, which caused some hard feelings.

Mississippi agencies are being taught to seal air leaks effectively in the houses of their low-income clients. But while reducing air leakage brings energy savings, it can also exacerbate indoor air quality problems unless comprehensive health and safety procedures are implemented. These health and safety procedures include testing for CO gas leaks, and worst-case draft; the results may necessitate replacing unsafe combustion appliances. Beginning this program year, Mississippi is making LIHEAP funds available to weatherization providers to replace unsafe heating systems. Without this alternative funding source, the health and safety effort required by whole-house weatherization would not be possible.

Tennessee Weatherization

The lessons learned from the Mississippi pilot give Tennessee an opportunity to provide their agencies and contractors with comprehensive technical training at an affordable price. Tennessee recently submitted their energy audit procedures to DOE for reapproval, as DOE regulations require every five years. While Tennessee's overall energy audit procedures comply with section 440.21 of DOE regulations, the existing priority lists for single-family houses and mobile homes are out of date. The priority lists were developed before program regulations were last updated and, therefore, do not reflect the higher standards placed on energy audit procedures by the final rule published on December 8, 2000.

At Tennessee's request, D&R developed new priority lists for single-family houses and mobile homes that comply with current regulations. Technical training will be required for local agencies and their contractors to fully adopt the whole-house approach to weatherization described in the new priority lists.

NRCERT and D&R will bring the successfully piloted technical training to Tennessee. Phase 1 training consists of a two- day introductory sessions for state and local weatherization staff in two different locations in the state, and three four-day regional training sessions—consisting of one day in the classroom, a second and third day mostly in the field, and a fourth half-day wrap-up in the classroom. The regional training is repeated in each of three locations across the state. The introductory session helps to put in context, for state and local agency staff, the technical training that will occur and the possible program design changes that will need to be made.

Introduction to Training

Two to three weeks prior to training, trainers will communicate to Tennessee staff the types of house that are good candidates for in-field training. Tennessee will select three to five candidate houses for in-field training; will e-mail information on, and pictures of, the houses to trainers; and will make sure that a meeting room for classroom training is reserved, transportation for in-field training has been arranged, classroom and in-field lunch arrangements have been made, and portable toilets have been arranged for the in-field locations.

One day prior to the introductory sessions, trainers will meet with state and local staff, visit the training house prospects, and select one primary and one back-up house. They will also set up audiovisual equipment and training props in the meeting room.

The introductory sessions consist of one half day in the classroom and one half day in the field. The half day of classroom instruction will answer the question, Why are we here? Trainers will present a brief background of DOE's Hot Climate Initiative and the Mississippi pilot training project. NRCERT will recount the experiences of the Virginia Weatherization program in bringing whole-house weatherization to Virginia. They will talk about their program evaluation in 1991, and describe the subsequent changes in how local agencies and contractors are paid for the completed weatherization work. They also describe how the Virginia Weatherization program adopted advanced air sealing, health, safety, diagnostic, and installation technologies and techniques. The reduction in client energy bills resulting from this transition will be highlighted.

The session will continue with an introduction to combustion appliance safety. This is an abbreviated version of what is covered in more detail, and for the benefit of subcontractors, in the regional training. The purpose of the introduction is to make the state and local staff aware of the significance of the subject, and to give those staff members the opportunity to discuss potential program design changes that might be necessary to ensure that combustion appliance safety is implemented. The trainers will discuss how to assess the safety of combustion appliances by measuring the levels of CO in the flue, as well as in ambient air. Alternatives to potentially dangerous unvented space heaters will be described. Participants will learn how to evaluate venting systems for effectiveness, safety, and code compliance.

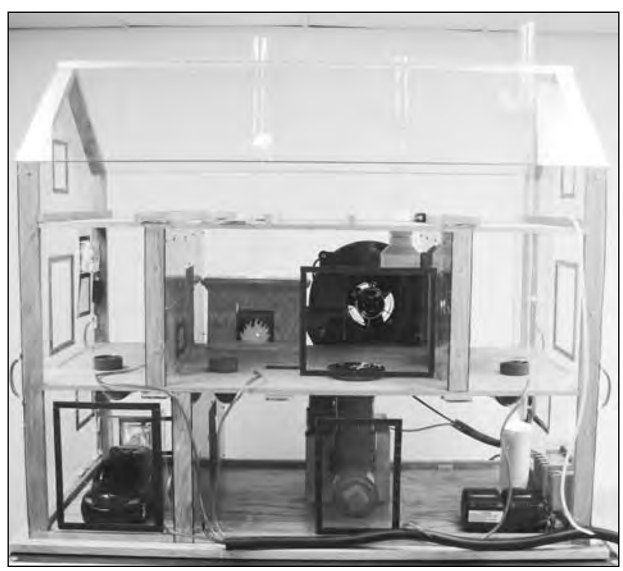
NRCERT will then use their House of Pressure to introduce and demonstrate blower-door-directed air sealing and zonal pressure diagnostics. (See "A Training House You Can Carry in Your Pickup.") This is an abbreviated version of what is covered in more detail, and also for the benefit of subcontractors, in the regional training. Reinforcing the combustion appliance

A Training House You Can Carry in Your Pickup

NRCERT developed the House of Pressure to complement its training activities. This training aid is used to visually demonstrate pressure and air flow dynamics within a residence. The 2 ft x 2 ft x 4 ft model has the ability to create and control air flow with working scaled reproductions of mechanical air distribution systems that are found in most homes. There are smoke generators in the water heater and in the fireplace to dramatically show the dangers of backdrafting, and there is a smoke generator in an exhaust pipe in the garage to show the danger of CO infiltration into conditioned space from a garage.

An automated performance testing (APT) device measures the pressures in eight different locations in the model house. Excel software then projects those pressures onto an LCD screen, showing pressure levels and the direction of air flow in every room of the house. The use of the House of Pressure can greatly reduce the amount of lecture time required to describe the information presented, and the training tool can be purchased by others to be used for educational purposes.

For more information, view the House of Pressure presentation on the Web at www.nrcert.org, or e-mail Anthony Cox of NRCERT at acox@chpc2.org, or acox990@aol.com.



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safety lessons, the trainers will use the House of Pressure to show how to put a house in the worst-case draft condition, so that proper drafting of combustion gases may be verified through measurement.

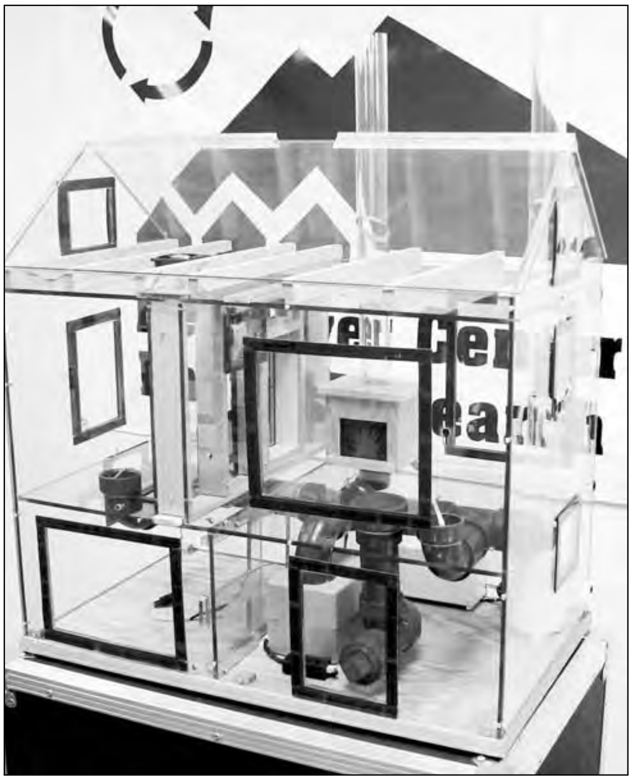
Training moves to the field for the remaining half day of the introductory session. At a weatherization-eligible house arranged by the host agency, the focus is on field estimation. The participants will be shown how to use the blower door and conduct pressure diagnostic testing. They will inspect the attic, hear about bypass-sealing techniques, and determine the adequacy of attic ventilation. They will be shown how to test the water heater and the stove for CO, and how to determine if the water heater is venting properly. Participants will be shown how to detect gas leaks, inspect the heating system, check the air conditioner, inspect the sidewalls, and learn sidewall insulation techniques. Time permitting, the potential of electric baseload measures will be evaluated by monitoring refrigerator energy use and assessing lighting conversion.

Regional Training

Preparations for the regional training will be the same as for the introduction. This includes finding appropriate houses for field training, arranging classroom space, and so on.

The advanced diagnostic and air sealing techniques to be taught on the first day address eliminating wasteful air infiltration. While there are many benefits to minimizing air leakage, there are also potential dangers. A furnace that spills CO into the home may harm the occupants; reducing air infiltration means that there will now be less fresh air to dilute the CO produced by the unsafe furnace. The trainers will demonstrate how to assess the safety of combustion appliances by measuring the levels of CO in the flue as well as in ambient air. Alternatives to potentially dangerous unvented space heaters will be described. Participants will learn how to evaluate venting systems for effectiveness, safety, and code compliance.

NRCERT will use their House of Pressure to demonstrate blower-door-directed air sealing and zonal pressure diagnostics. As in the introductory session, the trainers will use the House of Pressure to show how to put a house in the worst-case draft condition, so that proper drafting of combustion gases may be verified through measurement.



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NRCERT's House of Pressure visually demonstrates pressure and air flow dynamics within a house.

Building on the combustion appliance safety and pressure diagnostic training modules, participants will review how heat moves through a house, so that they can more fully understand effective insulation practices. Trainers will show participants practical attic insulation and sidewall dense-packing techniques that effectively reduce heat flow, in order to drive home theory. Diagrams and photographs will clearly illustrate critical junction points so that the participants will be able to identify them in the field. This leads to the next training module.

For the Mississippi pilot training, the Virginia field estimation form was modified to reflect housing stock and heating systems found in Mississippi. The trainers will use this form to walk participants through the field estimation, or house assessment, process. As they go through the form, participants will learn what data must be collected, what tests must be conducted, and what observations must be noted to comprehensively assess a home for weatherization. Any changes that would make the forms more appropriate to Tennessee's weatherization program will be solicited from the participants. Proper use of Tennessee's new single-family and mobile home priority lists will be reviewed.

Following the day of classroom instruction, training moves to the field for the next two days. As in the introductory training, the first in-field day will focus on field estimation. The first field day will conclude with agency staff and their contractors using training props to practice dense-packing sidewalls and sealing attic bypasses. On the second day in the field, participants will go into full production mode and complete the weatherization job. With trainer guidance, the participants will complete dense-packing the sidewalls with insulation, prepare the attic, seal bypasses, and insulate the attic. Any needed heating system work will be completed. Final blower door testing and safety checks will ensure that the house is left in a safe condition upon completion of weatherization work. Finally, trainers will stress the importance of cleaning up the job site, inside and out.

Back in the classroom on the final half day, the trainers will review the work done on the house and share the test results. They will discuss equipment setup and maintenance and address infrastructure and logistic issues, including equipment needs, the trucks or trailers needed to transport the equipment, and the possible sharing of equipment between agencies.

Trainers and Facilitators

NRCERT will provide classroom and in-field training. Anthony Cox from NRCERT and John Langford, a weatherization contractor from Lynchburg, Virginia, will lead hands-on technical training. The founder and owner of J&J Weatherization, Langford is particularly helpful in showing agencies and contractors how to set up, operate, and maintain equipment, and in providing a contractor's perspective. Alex Moore of D&R International will train agencies on the proper use of the priority lists, refrigerator replacement, and incandescent-to-fluorescent lighting conversion. The Virginia trainers from NRCERT and J&J were a powerful addition to the Mississippi pilot.

Moore, a professional engineer with D&R, provides technical support to DOE's Weatherization program headquarters staff in Washington, D.C. For DOE, he reviews the energy audit procedures that states use to evaluate what weatherization measures are most appropriate for a given house. In this review capacity, he developed new priority lists for use on Tennessee's single-family and mobile homes.

Tools and Equipment

NRCERT and J&J Weatherization will bring a truck and trailer to each regional training session containing all the tools and equipment needed to assess and weatherize the training site house. Tools include an insulation blower; hoses, wall tubes, and associated hardware; a13-kW generator; a blower door; 4–6 digital manometers; combustion analyzers; the House of Pressure; a heavy-duty, right-angle drill for drilling sidewalls; 3–4 cordless drills; circular saws; hand tools, including hammers, utility knives, and screwdrivers; flashlights and spotlights; and audiovisual equipment, including a laptop computer, LCD projector, screen, speakers, laser pointers, camcorder, and a printer. Since the training site house will be fully weatherized at the completion of the training session, it can be reported to DOE as a completion. Therefore, the host agency will be responsible for providing the weatherization materials such as cellulose insulation.


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Along with all the equipment needed to assess and weatherize a house, NRCERT provides material that may be hard to get locally, such as HEPA vacuums and other lead dust control equipment.

The complete set of tools and equipment that NRCERT will bring is valued at more than $35,000. NRCERT also, initially, provides materials that may be new or difficult to acquire locally, or that might otherwise require time to procure. Examples include HEPA vacuums and associated lead dust equipment (see "Lead Safe Weatherization," *HE* May/June '03, p. 24). Each participant will be provided a comprehensive training notebook containing copies of all the presentation materials, technical articles, field estimation forms, and a wealth of other technical resources that cover everything from the principles of heat movement to client education. A series of technical briefs that reinforce the training topics will be prepared and mailed to participants every few weeks following each regional training session.

There Is More Work to Do

Redirecting a state weatherization program requires more than technical training, since substantial investment in new equipment is required. Mississippi was able to tap LIHEAP funds to purchase the required hand tools and testing equipment for every agency. Purchase of the large insulation blowers and generators is being done in phases. Large trucks or trailers are needed to transport heavy insulation blowers, generators, and bulky bags of cellulose to the job site. Should Mississippi purchase the equipment for agencies to lend to contractors? What incentives can Mississippi provide to encourage private weatherization contractors to invest in large, expensive equipment? How will Tennessee, North Carolina, and Alabama meet these same challenges?

Some agencies may be hesitant to change their weatherization practices. Concerns about finding or keeping contractors who are able to conduct pressure diagnostics or densepack sidewalls with cellulose may keep agencies from embracing a whole-house approach, particularly if they weatherize just a few houses a year. DOE regulations have provisions that states must follow in effecting agency change. Agency contract and consolidation issues may also need to be addressed.

These infrastructure issues may fundamentally change the way agencies and contractors do business. Mississippi will face continuing challenges in building the infrastructure needed for whole-house approach weatherization to be adopted statewide. Tennessee, North Carolina, and Alabama will probably face the same questions as they develop their weatherization programs. Maybe these states will end up teaching Mississippi a thing or two.

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